

Schedule of Chilled Water Pump

Description: CHWP-CP-BF-01 TO 04

Description	Specified	Offered
Location	B2/F Chiller Plant Room	B2/F Chiller Plant Room
Area Served		For Tower 1 & 2
Numbers of units	4	4
Type	Horizontal Split Casing / Multi-Stage Centrifugal Type	Horizontal Split Casing
Manufacturer	-	Ebara
Country of Origin	-	Indonesia
Model Number	-	300x200 CNHA
Casing Material	Cast Iron	Cast Iron
Impeller Material	Stainless Steel SS316	Stainless Steel SS316
Shaft Material	Stainless Steel SS316	Stainless Steel SS316
Sleeves Material	Stainless Steel SS316	Stainless Steel SS316
Casing Ring Material	Stainless Steel SS316	Stainless Steel SS316
Shaft Seal	-	Mechanical Seal
Type of bearing	-	Ball Bearing
Suction connection size (mm)	-	300
Discharge connection size (mm)	-	200

(1) Pump		
Rated Working Pressure (kPa)		1600
Total Head (m)	46	46
Water Flow (l/s)	149	149
No. of Stage	-	1
Maximum Speed (rpm)	1450	1450
Type of Drive	-	Direct Drive
Brake Power (kW)	-	83.0
Minimum Pump Efficiency (%) at Operating Point	-	81



Schedule of Chilled Water Pump

Description: CHWP-CP-BF-01 TO 04

(2) Motor		
Manufacturer	-	Elektrim
Origin	-	Singapore
Type	-	TEFC IP55
Maximum Speed (rpm)	1450	1450
Output Power (kW)	-	110
Power Supply (V/Ph/Hz)	-	380/3/50
Starting Method	VSD	VSD
Operating Weight (kg)	-	2030.4
Overall Dimension (LxWxH) (mm)	-	1997 x 830 x 967



Description	Specified	Offered
Description	CHWP-CP-BF-05 to 06	
Location	B2/F Chiller Plant Room	B2/F Chiller Plant Room
Area Served		For Tower 1 & 2
Numbers of units	2	2
Type	Horizontal Split Casing / Multi-Stage Centrifugal Type	Horizontal Split Casing
Manufacturer	-	Ebara
Country of Origin	-	Indonesia
Model Number	-	200x100CNJA
Casing Material	Cast Iron	Cast Iron
Impeller Material	Stainless Steel SS316	Stainless Steel Grade 316
Shaft Material	Stainless Steel SS316	Stainless Steel Grade 316
Sleeves Material	Stainless Steel SS316	Stainless Steel Grade 316
Casing Ring Material	Stainless Steel SS316	Stainless Steel Grade 316
Shaft Seal	-	Mechanical Seal
Type of bearing	-	<u>Ball Bearing</u>
Suction connection size (mm)	-	200
Discharge connection size (mm)	-	100

(1) Pump		
Rated Working Pressure (kPa)		1600
Total Head (m)	46	46
Water Flow (l/s)	76	76
No. of Stage	-	1
Maximum Speed (rpm)	1450	1450
Type of Drive	-	Direct Drive
Brake Power (kW)	-	45.7
Minimum Pump Efficiency (%) at Operating Point	-	75

(2) Motor		
Manufacturer	-	ELEKTRIM
Origin	-	Singapore
Type	-	TEFC IP55
Maximum Speed (rpm)	1450	1450
Output Power (kW)	-	55
Power Supply (V/Ph/Hz)	-	380/3/50
Starting Method	VSD	VSD
Operating Weight (kg)	-	1240.8
Overall Dimension (LxWxH) (mm)	-	1676x755x874

Schedule of Hot Water Pump

Description: HWP-CP-BF-01 TO 02

Description	Specified	Offered
Location	B2/F Chiller Plant Room	B2/F Chiller Plant Room
Area Served		For Tower 1 & 2
Numbers of units	2	2
Type	Horizontal Split Casing / Multi-Stage Centrifugal Type	Horizontal Split Casing
Manufacturer	-	Ebara
Country of Origin	-	Indonesia
Model Number	-	200x150CNHA
Casing Material	Cast Iron	Cast Iron
Impeller Material	Stainless Steel SS316	Stainless Steel Grade 316
Shaft Material	Stainless Steel SS316	Stainless Steel Grade 316
Sleeves Material	Stainless Steel SS316	Stainless Steel Grade 316
Casing Ring Material	Stainless Steel SS316	Stainless Steel Grade 316
Shaft Seal	-	Mechanical Seal
Type of bearing	-	Ball Bearing
Suction connection size (mm)	-	200
Discharge connection size (mm)	-	150

(1) Pump		
Rated Working Pressure (kPa)		1600
Total Head (m)	37	37
Water Flow (l/s)	72.6	72.6
No. of Stage	-	1
Maximum Speed (rpm)	1450	1450
Type of Drive	-	Direct Drive
Brake Power (kW)	-	32.9
Minimum Pump Efficiency (%) at Operating Point	-	80

Schedule of Hot Water Pump

Description: HWP-CP-BF-01 TO 02

(2) Motor		
Manufacturer	-	ELEKTRIM
Origin	-	Singapore
Type	-	TEFC IP55
Maximum Speed (rpm)	1450	1450
Output Power (kW)	-	45
Power Supply (V/Ph/Hz)	-	380/3/50
Starting Method	VSD	
Operating Weight (kg)	-	977
Overall Dimension (LxWxH) (mm)	-	1526x670x774



Schedule of Chilled Water Pump

Description: CWP-CP-BF-01 TO 04

Description	Specified	Offered
Location	B2/F Chiller Plant Room	B2/F Chiller Plant Room
Area Served		For Tower 1 & 2
Numbers of units	4	4
Type	Horizontal Split Casing / Multi-Stage Centrifugal Type	Horizontal Split Casing
Manufacturer	-	Ebara
Country of Origin	-	Indonesia
Model Number	-	300x250 CNHA
Casing Material	Cast Iron	Cast Iron
Impeller Material	Stainless Steel SS316	Stainless Steel SS316
Shaft Material	Stainless Steel SS316	Stainless Steel SS316
Sleeves Material	Stainless Steel SS316	Stainless Steel SS316
Casing Ring Material	Stainless Steel SS316	Stainless Steel SS316
Shaft Seal	-	Mechanical Seal
Type of bearing	-	Ball Bearing
Suction connection size (mm)	-	300
Discharge connection size (mm)	-	250

(1) Pump		
Rated Working Pressure (kPa)		1600
Total Head (m)	46	37
Water Flow (l/s)	149	193
No. of Stage	-	1
Maximum Speed (rpm)	1450	1450
Type of Drive	-	Direct Drive
Brake Power (kW)	-	84.4
Minimum Pump Efficiency (%) at Operating Point	-	83



Schedule of Chilled Water Pump

Description: CWP-CP-BF-01 TO 04

(2) Motor		
Manufacturer	-	Elektrim
Origin	-	Singapore
Type	-	TEFC IP55
Maximum Speed (rpm)	1450	1450
Output Power (kW)	-	110
Power Supply (V/Ph/Hz)	-	380/3/50
Starting Method	VSD	VSD
Operating Weight (kg)	-	2018
Overall Dimension (LxWxH) (mm)	-	2007 x 900 x 997



Description	Specified	Offered
Description	CWP-CP-BF-05 to 06	
Location	B2/F Chiller Plant Room	B2/F Chiller Plant Room
Area Served		For Tower 1 & 2
Numbers of units	2	2
Type	Horizontal Split Casing / Multi-Stage Centrifugal Type	Horizontal Split Casing
Manufacturer	-	Ebara
Country of Origin	-	Indonesia
Model Number	-	250x150CNHA
Casing Material	Cast Iron	Cast Iron
Impeller Material	Stainless Steel SS316	Stainless Steel Grade 316
Shaft Material	Stainless Steel SS316	Stainless Steel Grade 316
Sleeves Material	Stainless Steel SS316	Stainless Steel Grade 316
Casing Ring Material	Stainless Steel SS316	Stainless Steel Grade 316
Shaft Seal	-	Mechanical Seal
Type of bearing	-	Ball Bearing
Suction connection size (mm)	-	250
Discharge connection size (mm)	-	150

(1) Pump		
Rated Working Pressure (kPa)		1600
Total Head (m)	37	37
Water Flow (l/s)	99	99
No. of Stage	-	1
Maximum Speed (rpm)	1450	1450
Type of Drive	-	Direct Drive
Brake Power (kW)	-	44.4
Minimum Pump Efficiency (%) at Operating Point	-	81

(2) Motor		
Manufacturer	-	ELEKTRIM
Origin	-	Singapore
Type	-	TEFC IP55
Maximum Speed (rpm)	1450	1450
Output Power (kW)	-	55
Power Supply (V/Ph/Hz)	-	380/3/50
Starting Method	VSD	VSD
Operating Weight (kg)	-	1310.4
Overall Dimension (LxWxH) (mm)	-	1686x755x864



Schedule of Source Water Pump

Description: SCP-CP-BF-01 TO 02

Description	Specified	Offered
Location	B2/F	B2/F
Area Served	For Heat Pump Plant	For Heat Pump Plant
Numbers of units	2	2
Type	Horizontal Split Casing / Multi-Stage Centrifugal Type	Horizontal Split Casing
Manufacturer	-	Ebara
Country of Origin	-	Indonesia
Model Number	-	150x150 CNFA
Casing Material	Cast Iron	Cast Iron
Impeller Material	Stainless Steel SS316	Stainless Steel Grade 316
Shaft Material	Stainless Steel SS316	Stainless Steel Grade 316
Sleeves Material	Stainless Steel SS316	Stainless Steel Grade 316
Casing Ring Material	Stainless Steel SS316	Stainless Steel Grade 316
Shaft Seal	-	Mechanical Seal
Type of bearing	-	Ball Bearing
Suction connection size (mm)	-	150
Discharge connection size (mm)	-	150

(1) Pump		
Rated Working Pressure (kPa)		1600
Total Head (m)	12.5	12.5
Water Flow (l/s)	51	51
No. of Stage	-	1
Maximum Speed (rpm)	1450	1450
Type of Drive	-	Direct Drive
Brake Power (kW)	-	7.8
Minimum Pump Efficiency (%) at Operating Point	-	80

Schedule of Source Water Pump

Description: SCP-CP-BF-01 TO 02

(2) Motor		
Manufacturer	-	Elektrim
Origin	-	Singapore
Type	-	TEFC IP55
Maximum Speed (rpm)	1450	1450
Output Power (kW)	-	11
Power Supply (V/Ph/Hz)	-	380/3/50
Starting Method	VSD	VSD
Operating Weight (kg)	-	532
Overall Dimension (LxWxH) (mm)	-	1249 x 620 x 660

Description	Specified	Offered
Description	ACWTP-CP-BF-01 to 02	ACWTP-CP-BF-01/02
Location	B2/F Chiller Plant Room	B2/F Chiller Plant Room
Area Served		For Tower 1 & 2
Numbers of units	2	2
Type	Vertical Multistage	Vertical Multistage
Manufacturer	-	Ebara
Country of Origin	-	Indonesia
Model Number	-	EVMK 50/9
Casing Material	Cast Iron	Cast Iron
Impeller Material	Stainless Steel SS316	Stainless Steel Grade 316
Shaft Material	Stainless Steel SS316	Stainless Steel Grade 316
Sleeves Material	Stainless Steel SS316	Stainless Steel Grade 316
Casing Ring Material	Stainless Steel SS316	Stainless Steel Grade 316
Shaft Seal	-	Mechanical Seal
Type of bearing	-	Ball Bearing
Suction connection size (mm)	-	65
Discharge connection size (mm)	-	50

(1) Pump		
Rated Working Pressure (kPa)		1600
Total Head (m)	98	98
Water Flow (l/s)	5.8	5.8
No. of Stage	-	9
Maximum Speed (rpm)	1450	1450
Type of Drive	-	Direct Drive
Brake Power (kW)	-	7.97
Minimum Pump Efficiency (%) at Operating Point	-	70

(2) Motor		
Manufacturer	-	ELEKTRIM
Origin	-	Singapore
Type	-	TEFC IP55
Maximum Speed (rpm)	1450	1450
Output Power (kW)	-	15
Power Supply (V/Ph/Hz)	-	380/3/50
Starting Method	VSD	VSD
Operating Weight (kg)	-	420
Overall Dimension (LxWxH) (mm)	-	350x350x1567

Schedule of Bleed Off Water Pump

Description: ACBOP-CP-RF-01 TO 02

Description	Specified	Offered
Location	R/F	R/F
Area Served	Bleed off water to PD Flushing Tank	Bleed off water to PD Flushing Tank
Numbers of units	2	2
Type	Vertical Multi-Stage	Vertical Multistage
Manufacturer	-	Ebara
Country of Origin	-	Indonesia
Model Number	-	EVMK 50/2
Casing Material	Cast Iron	Cast Iron
Impeller Material	Stainless Steel SS316	Stainless Steel SS316
Shaft Material	Stainless Steel SS316	Stainless Steel SS316
Sleeves Material	Stainless Steel SS316	Stainless Steel SS316
Casing Ring Material	Stainless Steel SS316	Stainless Steel SS316
Shaft Seal	-	Mechanical Seal
Type of bearing	-	Ball
Suction connection size (mm)	-	65
Discharge connection size (mm)	-	50

(1) Pump		
Rated Working Pressure (kPa)		3000
Total Head (m)	12	12
Water Flow (l/s)	5.8	5.8
No. of Stage	-	2
Maximum Speed (rpm)	1450	1450
Type of Drive	-	Direct
Brake Power (kW)	-	0.97
Minimum Pump Efficiency (%) at Operating Point	-	70

Schedule of Bleed Off Water Pump

Description: ACBOP-CP-RF-01 TO 02

(2) Motor		
Manufacturer	-	ELEKTRIM
Origin	-	Singapore
Type	-	TEFC IP55
Maximum Speed (rpm)	1450	1450
Output Power (kW)	-	4
Power Supply (V/Ph/Hz)	-	380/3/50
Starting Method	Star-Delta	DOL
Operating Weight (kg)	-	198
Overall Dimension (LxWxH) (mm)	-	350x350x739

Sound Power Level: 65dB(A)

Description	Specified	Offered
Description	SCWBP-CP-URF-01 to 02	SCWBP-CP-URF-01 to 02
Location	URF	URF
Area Served		For Tower 1 & 2
Numbers of units	2	2
Type	Vertical Multi-Stage	Vertical Multi-Stage
Manufacturer	-	Ebara
Country of Origin	-	Indonesia
Model Number	-	EVMK 40/3
Casing Material	Cast Iron	Cast Iron
Impeller Material	Stainless Steel SS316	Stainless Steel Grade 316
Shaft Material	Stainless Steel SS316	Stainless Steel Grade 316
Sleeves Material	Stainless Steel SS316	Stainless Steel Grade 316
Casing Ring Material	Stainless Steel SS316	Stainless Steel Grade 316
Shaft Seal	-	Mechanical Seal
Type of bearing	-	Ball Bearing
Suction connection size (mm)	-	50
Discharge connection size (mm)	-	40

(1) Pump		
Rated Working Pressure (kPa)		1600
Total Head (m)	15	15
Water Flow (l/s)	3.504	3.504
No. of Stage	-	3
Maximum Speed (rpm)	1450	1450
Type of Drive	-	Direct Drive
Brake Power (kW)	-	0.86
Minimum Pump Efficiency (%) at Operating Point	-	70



(2) Motor		
Manufacturer	-	ELEKTRIM
Origin	-	Singapore
Type	-	TEFC IP55
Maximum Speed (rpm)	1450	1450
Output Power (kW)	-	1.5
Power Supply (V/Ph/Hz)	-	380/3/50
Starting Method	-	Star Delta
Operating Weight (kg)	-	135
Overall Dimension (LxWxH) (mm)	-	175x175x875



Description	Specified	Offered
Description	CRP-CP-BF-01 to 02	CRP-CP-BF-01 to 02
Location	B2/F Chiller Plant Room	B2/F Chiller Plant Room
Area Served		For Tower 1 & 2
Numbers of units	2	2
Type	Vertical Multistage	Vertical Multistage
Manufacturer	-	Ebara
Country of Origin	-	Italy
Model Number	-	EVM 3 11F5/0.37
Casing Material	Stainless Steel SS316	Stainless Steel Grade 316
Impeller Material	Stainless Steel SS316	Stainless Steel Grade 316
Shaft Material	Stainless Steel SS316	Stainless Steel Grade 316
Sleeves Material	Stainless Steel SS316	Stainless Steel Grade 316
Casing Ring Material	Stainless Steel SS316	Stainless Steel Grade 316
Shaft Seal	-	Mechanical Seal
Type of bearing	-	Ball Bearing
Suction connection size (mm)	-	25
Discharge connection size (mm)	-	25

(1) Pump		
Rated Working Pressure (kPa)		1600
Total Head (m)	19	19
Water Flow (l/s)	0.18	0.18
No. of Stage	-	11
Maximum Speed (rpm)	1450	1450
Type of Drive	-	Direct Drive
Brake Power (kW)	-	0.06
Minimum Pump Efficiency (%) at Operating Point	-	60



(2) Motor		
Manufacturer	-	ELEKTRIM
Origin	-	Singapore
Type	-	TEFC IP55
Maximum Speed (rpm)	1450	1450
Output Power (kW)	-	0.37
Power Supply (V/Ph/Hz)	-	380/3/50
Starting Method		Star Delta
Operating Weight (kg)	-	32.4
Overall Dimension (LxWxH) (mm)	-	250x150x610



Description	Specified	Offered
Description	CRP-CP-BF-03 to 04	CRP-CP-BF-03 to 04
Location	B2/F Chiller Plant Room	B2/F Chiller Plant Room
Area Served		For Tower 1 & 2
Numbers of units	2	2
Type	Vertical Multistage	Vertical Multistage
Manufacturer	-	Ebara
Country of Origin	-	Italy
Model Number	-	EVM 3 11F5/0.37
Casing Material	Stainless Steel SS316	Stainless Steel Grade 316
Impeller Material	Stainless Steel SS316	Stainless Steel Grade 316
Shaft Material	Stainless Steel SS316	Stainless Steel Grade 316
Sleeves Material	Stainless Steel SS316	Stainless Steel Grade 316
Casing Ring Material	Stainless Steel SS316	Stainless Steel Grade 316
Shaft Seal	-	Mechanical Seal
Type of bearing	-	Ball Bearing
Suction connection size (mm)	-	25
Discharge connection size (mm)	-	25

(1) Pump		
Rated Working Pressure (kPa)		1600
Total Head (m)	16	16
Water Flow (l/s)	0.21	0.21
No. of Stage	-	1
Maximum Speed (rpm)	1450	1450
Type of Drive	-	Direct Drive
Brake Power (kW)	-	0.06
Minimum Pump Efficiency (%) at Operating Point	-	61



(2) Motor		
Manufacturer	-	ELEKTRIM
Origin	-	Singapore
Type	-	TEFC IP55
Maximum Speed (rpm)	1450	1450
Output Power (kW)	-	0.37
Power Supply (V/Ph/Hz)	-	380/3/50
Starting Method	VSD	Star Delta
Operating Weight (kg)	-	32.4
Overall Dimension (LxWxH) (mm)	-	250x150x610



Schedule of Feed & Expansion Pump

Description: FEWP-CP-URF-01 TO 02

Description	Specified	Offered
Location	UR/F CT Make-up Water Tank and F&E Tank Room	UR/F CT Make-up Water Tank and F&E Tank Room
Area Served	A/C Feed & Expansion Tank	A/C Feed & Expansion Tank
Numbers of units	2	2
Type	Vertical Multistage	Vertical Multistage
Manufacturer	-	Ebara
Country of Origin	-	Italy
Model Number	-	EVM 10 5F5/0.55
Casing Material	Stainless Steel SS316	Stainless Steel SS316
Impeller Material	Stainless Steel SS316	Stainless Steel SS316
Shaft Material	Stainless Steel SS316	Stainless Steel SS316
Sleeves Material	Stainless Steel SS316	Stainless Steel SS316
Casing Ring Material	Stainless Steel SS316	Stainless Steel SS316
Shaft Seal	-	Mechanical
Type of bearing	-	Ball
Suction connection size (mm)	-	40
Discharge connection size (mm)	-	40

(1) Pump		
Rated Working Pressure (kPa)		1600
Total Head (m)	10	10
Water Flow (l/s)	0.83	0.83
No. of Stage	-	5
Maximum Speed (rpm)	1450	1450
Type of Drive	-	Direct Drive
Brake Power (kW)	-	0.13
Minimum Pump Efficiency (%) at Operating Point	-	65



Schedule of Feed & Expansion Pump

Description: FEWP-CP-URF-01 TO 02

(2) Motor		
Manufacturer	-	ELEKTRIM
Origin	-	Singapore
Type	-	TEFC IP55
Maximum Speed (rpm)	1450	1450
Output Power (kW)	-	0.55
Power Supply (V/Ph/Hz)	-	380/3/50
Starting Method	DOL	DOL
Operating Weight (kg)	-	41
Overall Dimension (LxWxH) (mm)	-	280x130x677

